Optimizing Doctor Availability

A Project Report

Submitted by

Ms. Divya Manik Rathod
(Roll no. 2023MCA58)

Fulfilment for the award of the degree

MASTER OF COMPUTER APPLICATIONS

UNDER THE GUIDANCE OF
PROF. Asawari Sawant

MKSSS'S
K.B. JOSHI INSTITUTE OF INFORMATION TECHNOLOGY,
KARVE NAGAR, PUNE 52

AFFILIATED BY
SNDT WOMEN'S UNIVERSITY, MUMBAI.
2023-2025

CERTIFICATE

This is to certify that the project report titled

"Optimizing Doctor Availability"

Miss Divya Rathod **Roll no.2023MCA58**

Is a Bonafide work carried out by them under the supervision of Mrs. Asawari Sawant. Project is completed and approved for fulfilment of the requirement SNDT Women's University, Mumbai forth award of Degree of Master of Computer Application (MCA).

The project work was carried during the period 2023-25

Project Guide **Principal**

External Examiner **Student**

Sai Clinic

Dr. Satish Bhagwan Balwant

B.A.M.S, Regl. No.: 1/52610 A1

Yashodeep Chowk, Near Yash Mitra Mandal, Warje Malwadi, Pune - 411 058.

Mob. No. : 98813 71225 E-mail : drsatishbalwant@yahoo.com

Timing : 9:00am to 1:00pm - 5:00 to 10:00pm

No. Date : / /20

Certificate

This is to certify that Miss Divya Rathod has satisfactorily completed the project titled “Optimizing Doctor Availability” application under my guidance/ supervision for my clinic work. She has Analyzed our requirements properly and developed the working product within the stipulated time.

Dispenced by:
Name & Address of Medical Store:
Date of Dispencing:

Doctor Signature / Stamp

लक्ष्मी मेडीकल

शॉप नं.२, यश मित्र मंडळ चौक, वारजे

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ACKNOWLEDGEMENT

We have a great pleasure and satisfaction in presenting the project on “Optimizing Doctor Availability”. We would like to show our gratitude towards our Principal Dr. Anagha Vaidya for providing good facilities and valuable guidance.

We would like to express our gratitude to Prof. Asawari Sawant for their project guidance and support, there guidance throughout helped us in completion of project.

We would like to thank Mr. Girish kumthekar for showing faith in us and allowing us to develop software “Optimizing Doctor Availability” for their Canteen.

We are thankful for all teaching, non-teaching staff of the college who directly and indirectly helped us in completion of project.

INTRODUCTION

In today's fast-paced healthcare landscape, the efficient utilization of resources is paramount for ensuring timely and effective patient care. The project, "Optimizing Doctor Availability and Appointment Allocating in Hospitals through Digital Technology and AI Integration," addresses this critical need by leveraging cutting-edge digital technologies and artificial intelligence (AI) to streamline the scheduling and allocation processes in hospitals.

Traditional appointment systems often suffer from inefficiencies, leading to long waiting times, suboptimal resource allocation, and patient dissatisfaction. This project aims to revolutionize the healthcare system by introducing a comprehensive solution that harnesses the power of AI algorithms and digital tools. By integrating advanced technologies, the project seeks to optimize doctor availability, enhance appointment scheduling, and minimize patient waiting times.

The implementation of AI algorithms will enable hospitals to analyse historical data, patient preferences, and doctor availability patterns, facilitating a predictive and dynamic appointment system. This not only ensures that patients receive timely medical attention but also maximizes the utilization of healthcare professionals' time. Additionally, the incorporation of digital technology will provide a user-friendly interface for both healthcare providers and patients, fostering a seamless and efficient communication channel.

Ultimately, this project envisions a healthcare system where the intersection of digital technology and AI transforms the way appointments are managed, leading to improved patient experiences, enhanced resource utilization, and overall operational efficiency in hospitals.

COMPANY PROFILE

- **COMPANY NAME**

  Sai Clinic

- **OWNER NAME**

  Dr. Satish Bhagwan Balwant

- **ADDRESS**

  Yashodeep Chowk, Near Yash Mitra Mandal, Warje Malwadi,
  Pune – 411058.

- **CONTACT NO.**

  9881371225

EXISTING SYSTEM

- Currently, healthcare appointment systems often rely on manual processes and lack optimization features. Patients typically book appointments through phone calls or in-person visits, leading to inefficiencies in doctor availability and appointment allocation. Information accessibility is limited, hindering informed decision-making.
- This traditional approach lacks real-time data presentation and AI integration. The proposed project seeks to overcome these limitations by introducing a digital solution that streamlines the appointment process and leverages AI for efficient doctor-patient interactions and optimal resource utilization.

DRAWBACKS OF EXISTING SYSTEM

- Technological Barriers
- Initial Implementation Challenges
- Security Risks
- Cost Implications

PROPOSED SYSTEM

- The proposed system aims to revolutionize healthcare appointment management by leveraging digital technology and artificial intelligence.

- It comprises three modules: Hospital, Doctor, and Patient. The system minimizes doctor appointments and allocates optimal time slots, enhancing patient-doctor communication

- A user-friendly interface allows patients to search for doctors based on medical needs, providing comprehensive details for informed decision-making. Real-time appointment data, doctor approvals, and email notifications streamline the process, ensuring efficient resource utilization. By integrating AI, the system improves overall healthcare service delivery, offering a modern, efficient, and patient-centric approach to appointment scheduling and management.

PROJECT OBJECTIVE

- The project's primary objective is to optimize doctor availability and appointment allocation in hospitals through digital technology and AI integration.

- Specific goals include minimizing the number of appointments for each doctor, allocating minimum time slots for individual appointments, and enhancing patient-doctor communication.

- The project aims to provide a device-friendly interface for patients to search for doctors, offer comprehensive information for informed decision-making, and implement features such as real-time appointment data and email notifications, ultimately improving the efficiency and effectiveness of healthcare services.

SCOPE OF PROPOSED SYSTEM

- The scope of this project extends to revolutionizing healthcare appointment systems by utilizing digital technology and AI integration.
- It encompasses the development of a comprehensive platform with three main modules: Hospital, Doctor, and Patient.
- The system aims to minimize doctor appointments, allocate optimal time slots, and enhance patient-doctor communication.
- The device-friendly interface enables patients to easily search for doctors based on medical needs.
- The project's scope includes real-time data presentation, email notifications, and ultimately aims to improve overall healthcare service delivery by ensuring efficient resource utilization and streamlined appointment processes.

FEASIBILITY STUDY

The feasibility of the project is analysed in this phase and business proposal is put forth with a very general plan for the project and some cost estimates. During system analysis the feasibility study of the proposed system is to be carried out. This is to ensure that the proposed system is not a burden to the company. For feasibility analysis, some understanding of the major requirements for the system is essential.

Technical Feasibility:

- This study is carried out to check the technical feasibility, that is, the technical requirements of the system.
- Any system developed must not have a high demand on the available technical resources. This will lead to high demands on the available technical resources. This will lead to high demands being placed on the client.
- The developed system must have a modest requirement, as only minimal or null changes are required for implementing this system.

Economic Feasibility:

- This study is carried out to check the economic impact that the system will have on the organization. The amount of fund that the company can pour into the research and development of the system is limited.
- The expenditures must be justified. Thus, the developed system as well within the budget and this was achieved because most of the technologies used are freely available. Only the customized products had to be purchased.

Social Feasibility:

- The aspect of study is to check the level of acceptance of the system by the user. This includes the process of training the user to use the system efficiently.
- The user must not feel threatened by the system, instead must accept it as a necessity.
- The level of acceptance by the users solely depends on the methods that are employed to educate the user about the system and to make him familiar with it. His level of confidence must be raised so that he is also able to make some constructive criticism, which is welcomed, as he is the final user of the system.

Hardware & Software Requirements

HARDWARE REQUIREMENTS:

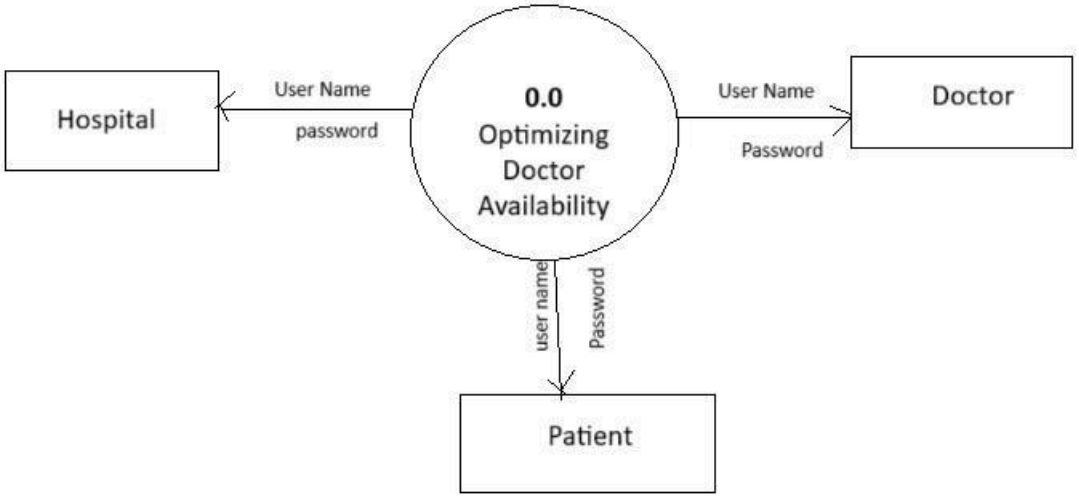
- Processor - I3/Intel Processor
- Hard Disk - 160GB
- Key Board - Standard Windows Keyboard
- Mouse - Two or Three Button Mouse
- Monitor - SVGA
- RAM - 8GB

SOFTWARE REQUIREMENTS:

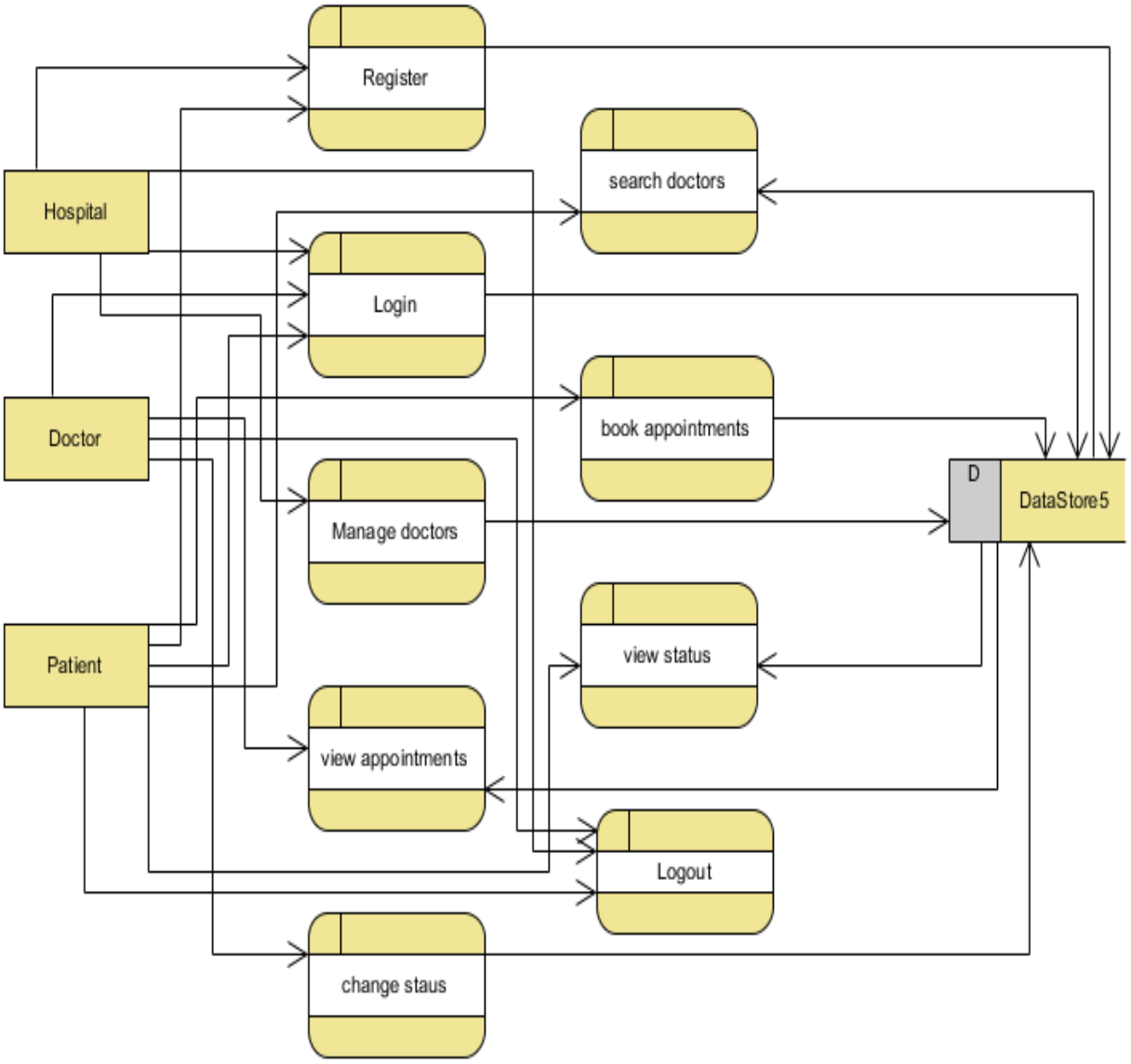
- Operating System : Windows 7/8/10
- Programming Language : Spring boot and React
- IDE/Workbench : IntelliJ and Visual Studio Code
- Database : MySQL Workbench

DIAGRAM

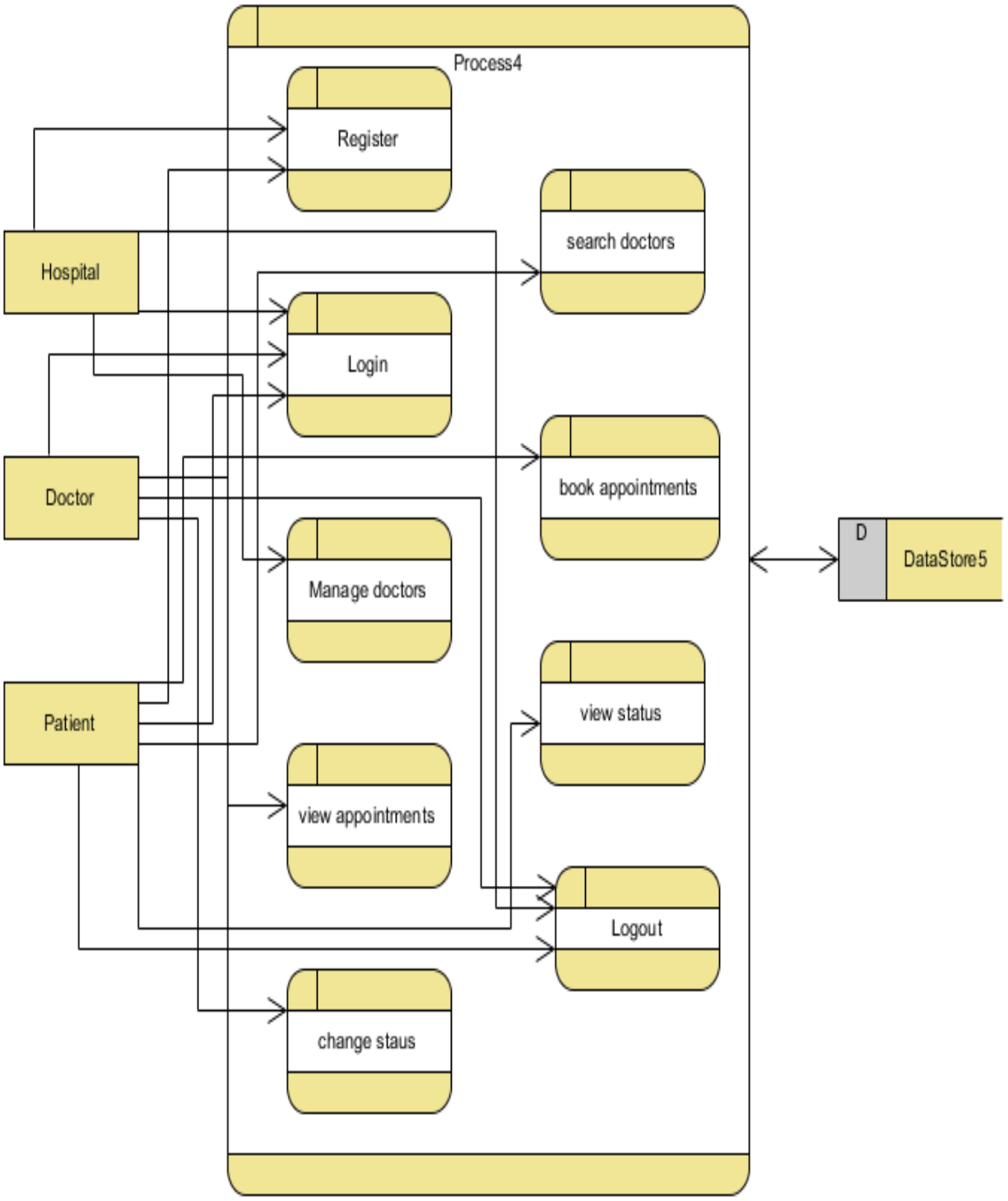
Context Level Diagram


Hospital
User Name
password
0.0
Optimizing
Doctor
Availability
User Name
Password
Doctor
user name
Password
Patient

DFD Diagram:

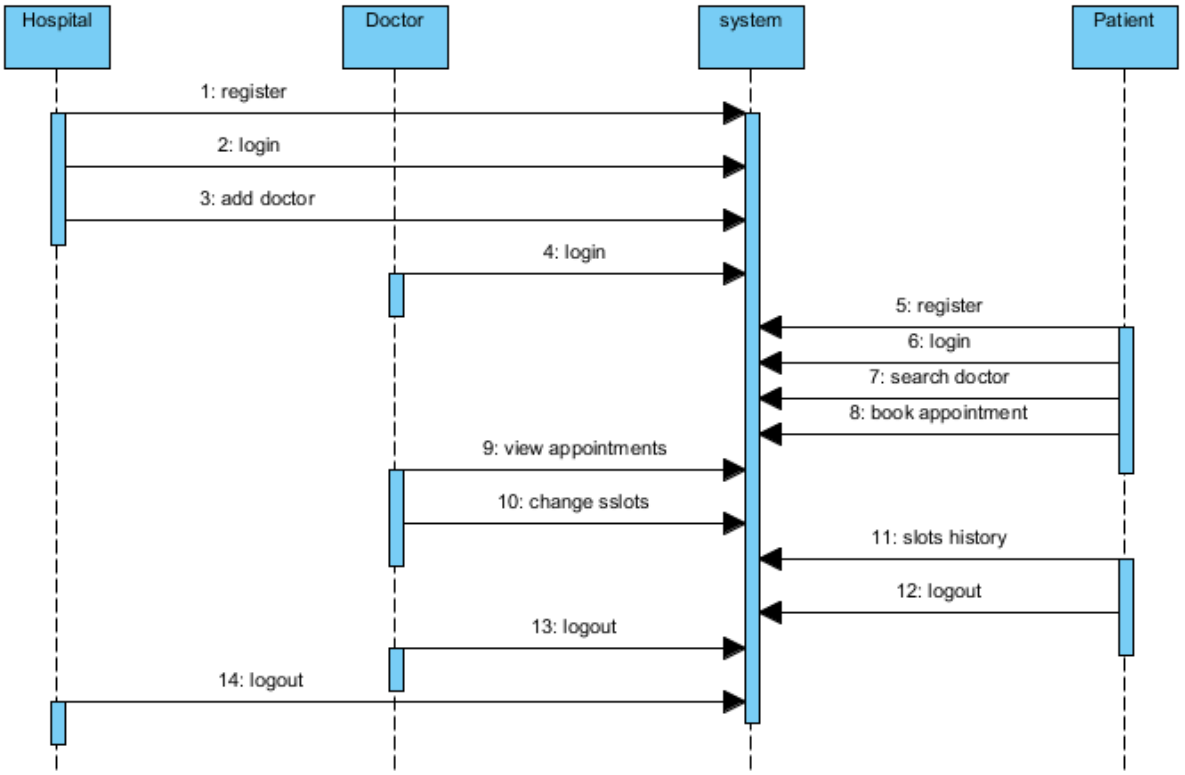

Register
search doctors
Hospital
Login
Doctor
book appointments
D
DataStore5
Manage doctors
view status
Patient
view appointments
Logout
change staus

Level – 1 DFD

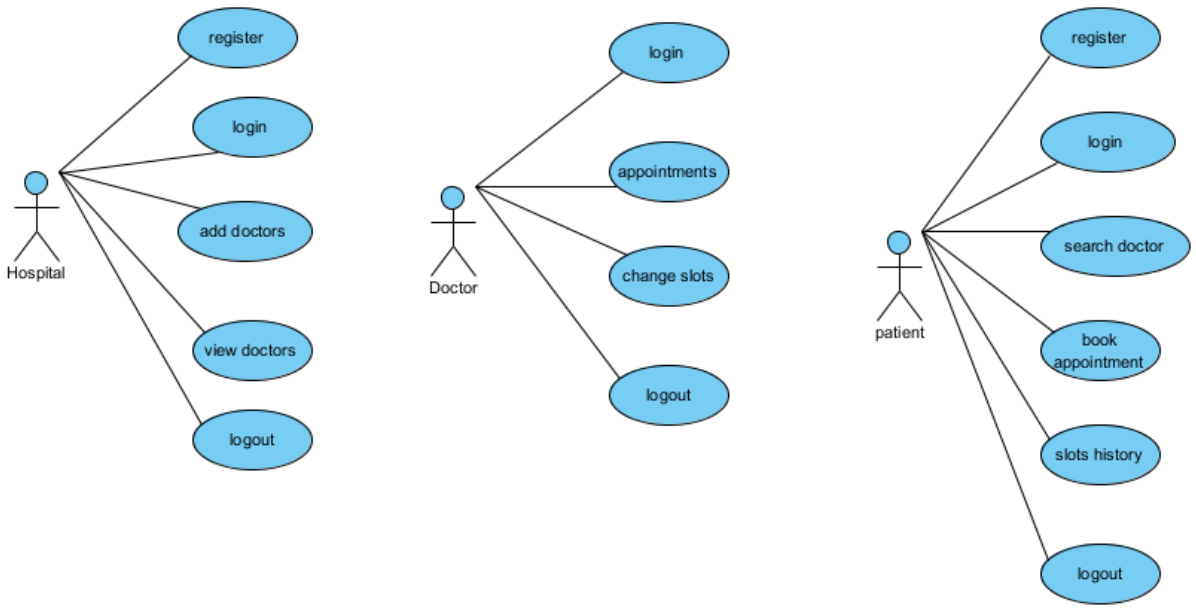

Process4
Register
search doctors
Hospital
Login
book appointments
Doctor
D
DataStore5
Manage doctors
view status
Patient
view appointments
Logout
change staus

Level-2 DFD

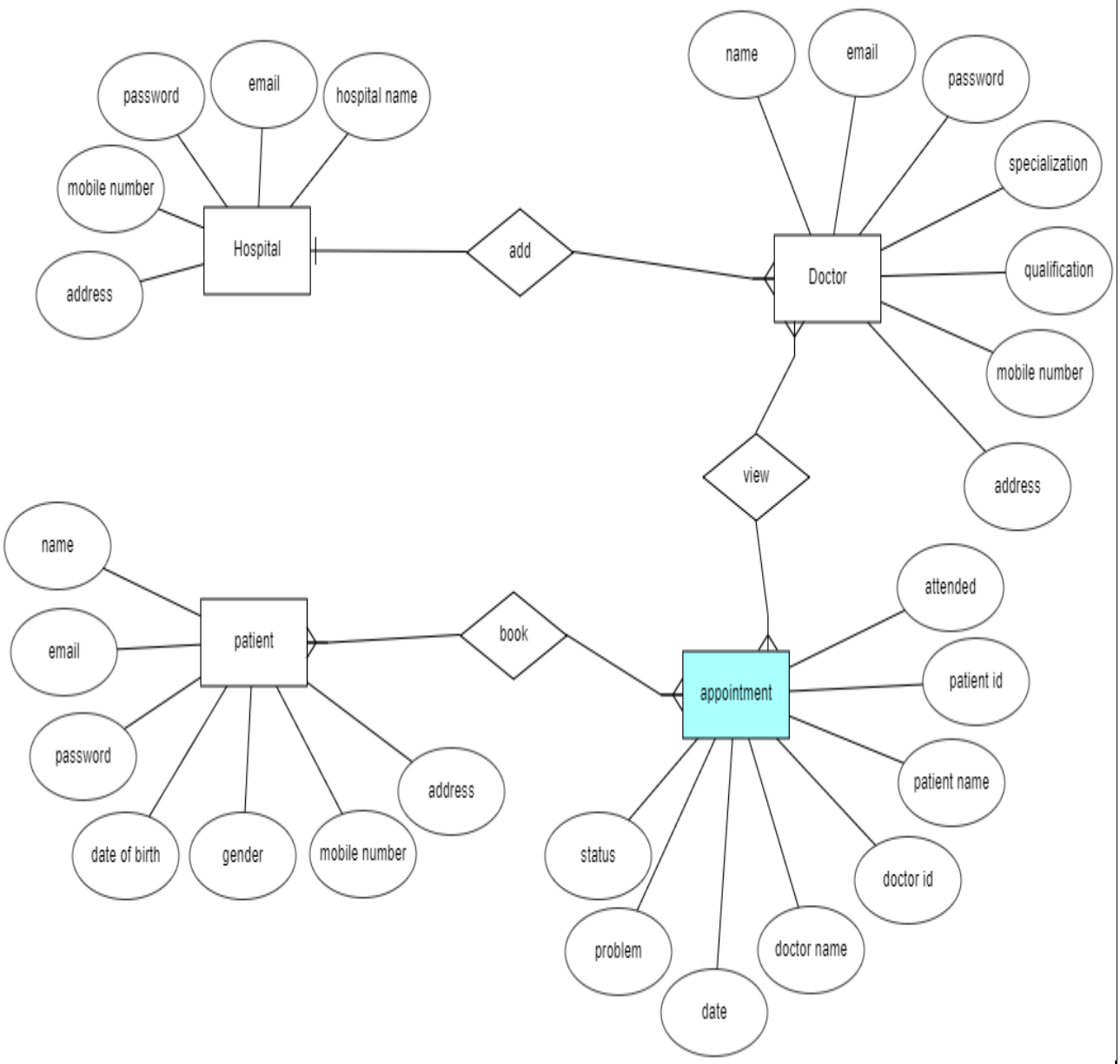
Sequence Diagram:



Use Case Diagram:



ER Diagram:


password
email
hospital name
mobile number
address
Hospital
add
name
email
password
specialization
qualification
mobile number
address
Doctor
view
name
email
password
date of birth
gender
mobile number
address
patient
book
appointment
attended
patient id
patient name
doctor id
doctor name
date
problem
status

FILE DESIGNING:

Appointment

Field	Data Type	Null	Key	Default	Constraints
id	bigint	No	PRI	NULL	Auto_increment
appointment_data	varchar (255)	Yes		NULL	
attented	bit(1)	No		NULL	
doctor_id	bigint	Yes		NULL	
doctor_name	varchar (255)	Yes		NULL	
patient_id	bigint	Yes		NULL	
patient_name	varchar (255)	Yes		NULL	
problem	varchar (255)	Yes		NULL	
status	enum('pending', 'scheduled', 'completed','cancel')	Yes		NULL	

Doctor

Field	Data Type	Null	Key	Default	Extra
id	bigint	No	PRI	NULL	Auto_increment
address	varchar (255)	Yes		NULL	
email	varchar (255)	Yes		NULL	
hospital_name	varchar (255)	Yes		NULL	
name	varchar (255)	Yes		NULL	
password	varchar (255)	Yes		NULL	
phone_number	varchar (255)	Yes		NULL	
qualification	varchar (255)	Yes		NULL	
specialization	varchar (255)	Yes		NULL	

Doctor_appointments

Field	Data Type	Null	Key	Default	Extra
doctor_id	bigint	No	MUL	NULL	
appointments_id	bigint	No	MUL	NULL	

Hospital

Field	Data Type	Null	Key	Default	Extra
id	bigint	No	PRI	NULL	Auto_increment
address	varchar (255)	Yes		NULL	
email	varchar (255)	Yes		NULL	
Password	varchar (255)	Yes		NULL	
Phone_number	varchar (255)	Yes		NULL	
username	varchar (255)	Yes		NULL	

Hospital_doctor

Field	Data Type	Null	Key	Default	Extra
hospital_id	bigint	No	MUL	NULL	
doctor_id	bigint	No	MUL	NULL	

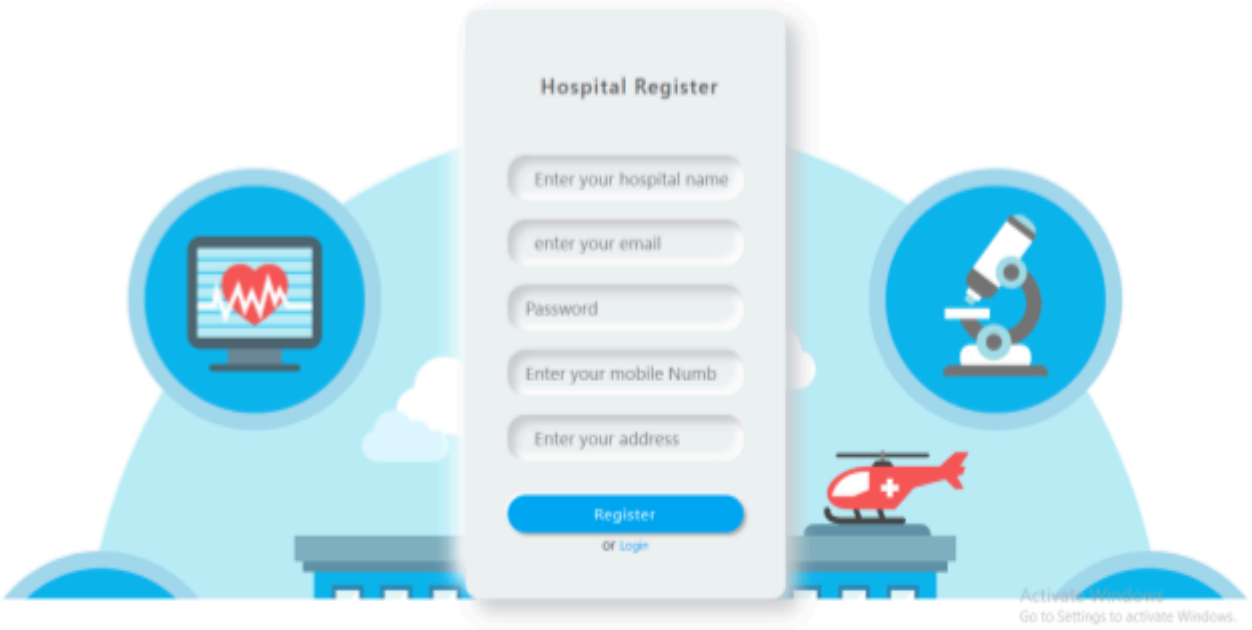
Patient

Field	Data Type	Null	Key	Default	Extra
id	bigint	No	PRI	NULL	Auto_increment
address	varchar (255)	Yes		NULL	
date_of_birth	varchar (255)	Yes		NULL	
email	varchar (255)	Yes		NULL	
gender	varchar (255)	Yes		NULL	
name	varchar (255)	Yes		NULL	
password	varchar (255)	Yes		NULL	
phone_number	varchar (255)	Yes		NULL	

Input Screens

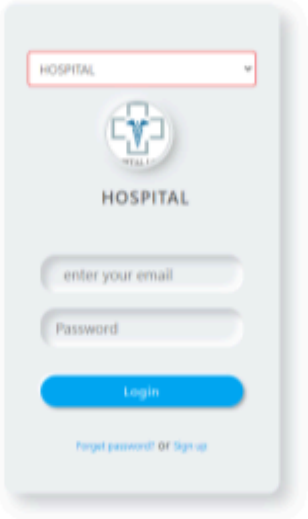
Home Page:

Hospital Register:



Hospital Login:

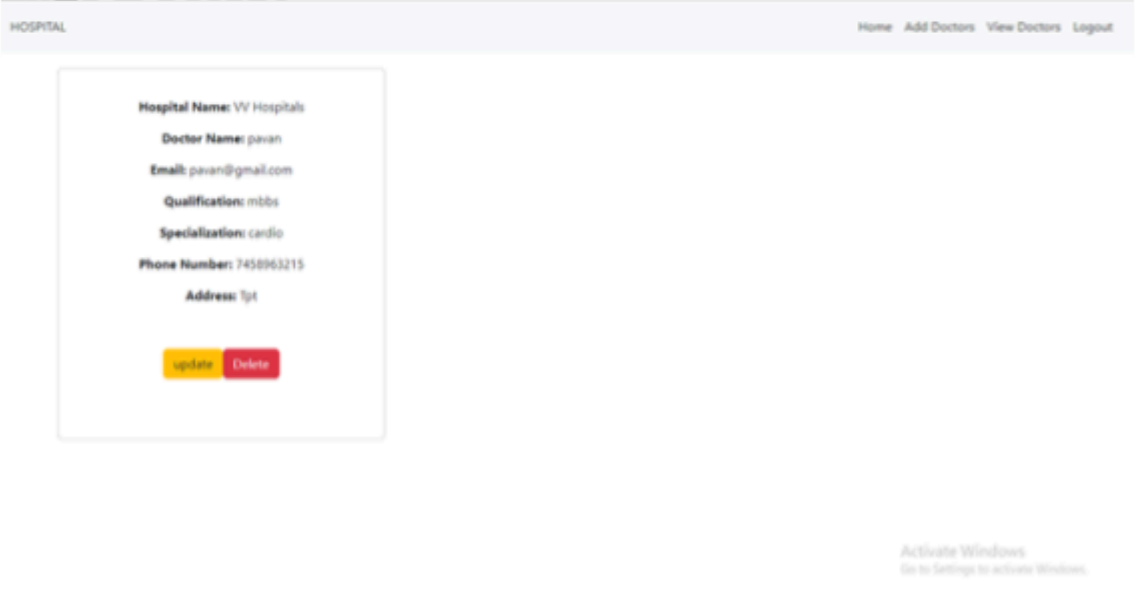
:



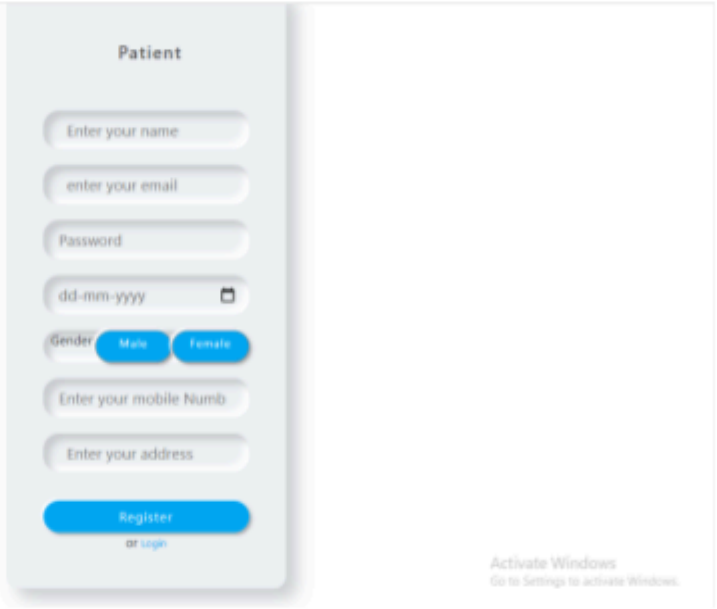
Hospital Home:

Add Doctor:

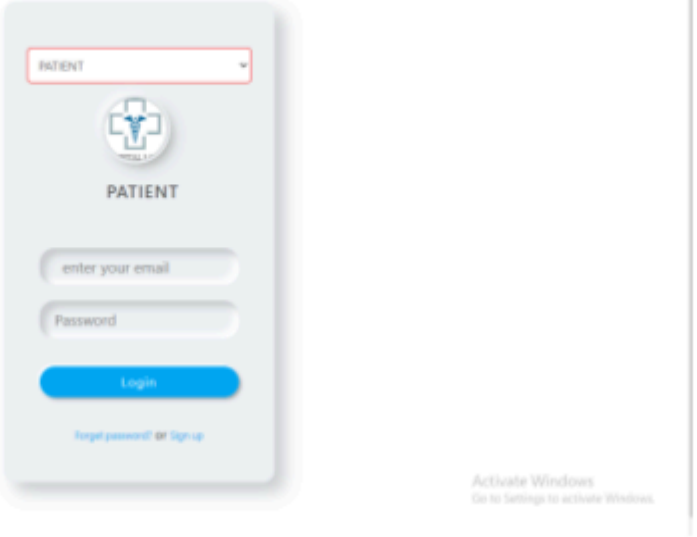
Manage Doctor:



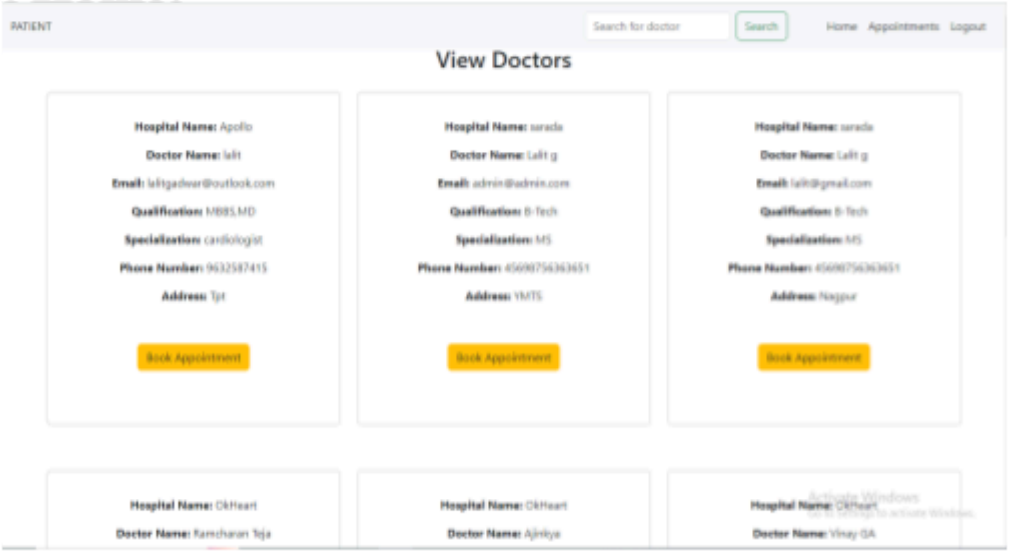
Patient Register:



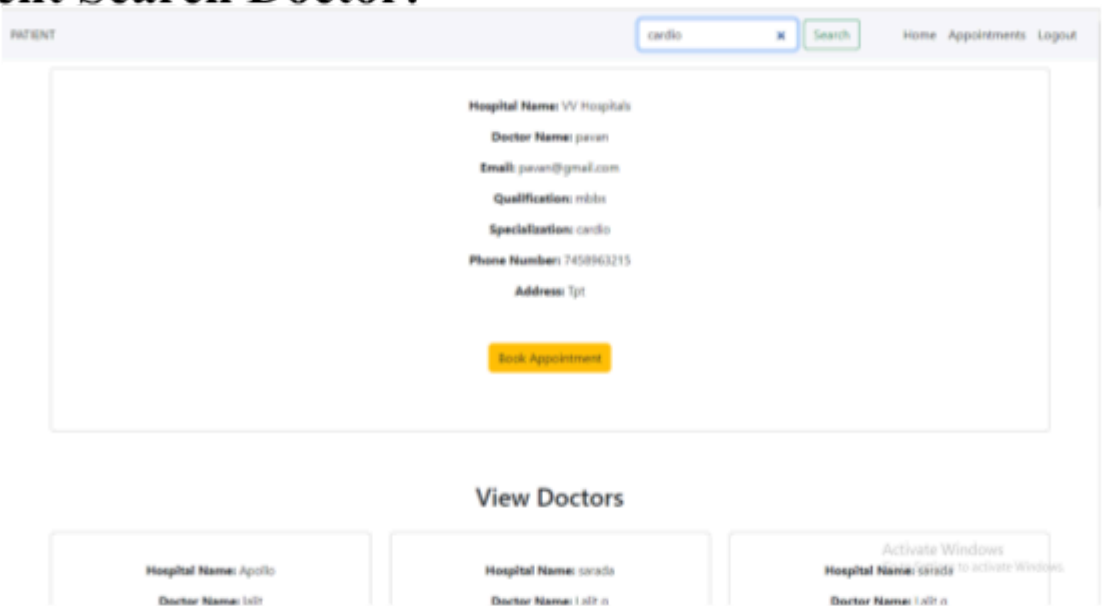
Patient Login: :



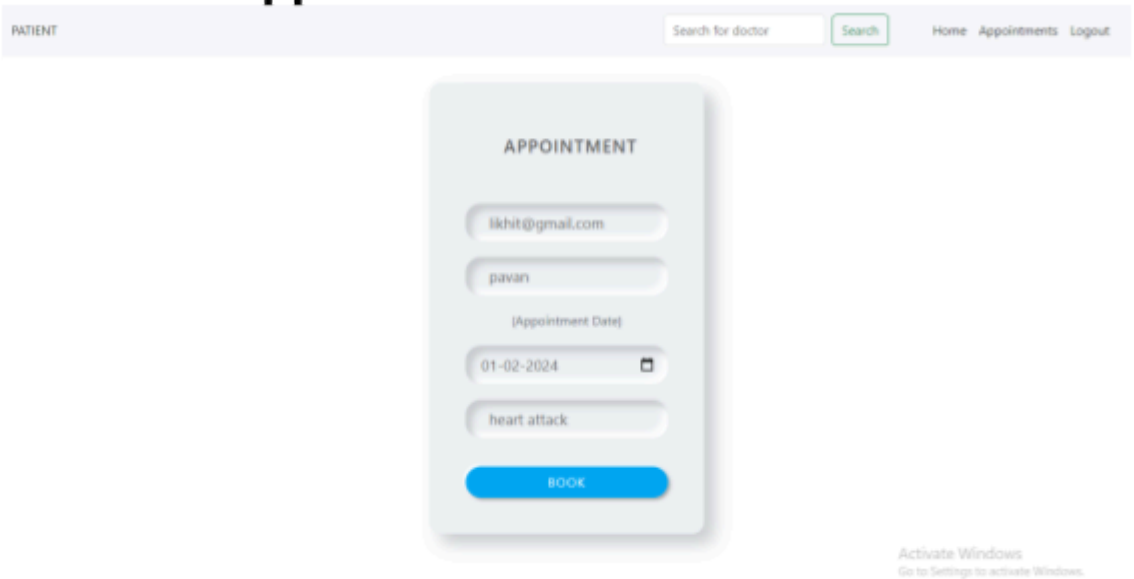
Patient Home:



Patient Search Doctor:



Patient Book Appointment:



Patient View Appointment:

All Appointments

#	Doctor Name	Health issues	Date	Status
1	lalit	diarrhoea	2024-02-02	
2	Rakesh	diarrhoea	2024-01-25	COMPLETED
3	Rakesh	diarrhoea	2024-01-25	COMPLETED
4	Rakesh	diarrhoea	2024-01-25	COMPLETED
5	pavan	heart attack	2024-02-01	

Patient view status of his appointment:

PATIENT

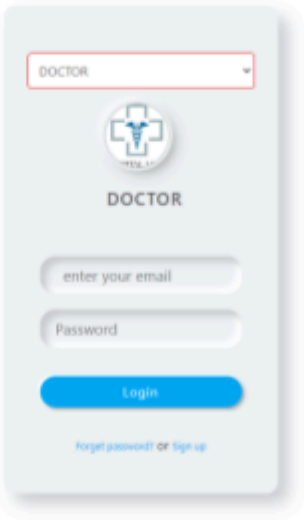
Search for doctor | Search | Home | Appointments | Logout

All Appointments

#	Doctor Name	Health Issues	Date	Status
1	lalit	diarrhoea	2024-02-02	
2	Rakesh	diarrhoea	2024-01-25	COMPLETED
3	Rakesh	diarrhoea	2024-01-25	COMPLETED
4	Rakesh	diarrhoea	2024-01-25	COMPLETED
5	pavan	heart attack	2024-02-01	COMPLETED

Activate Windows
Go to Settings to activate Windows.

Doctor Login:



Activate Windows
Go to Settings to activate Windows.

Doctor Home:

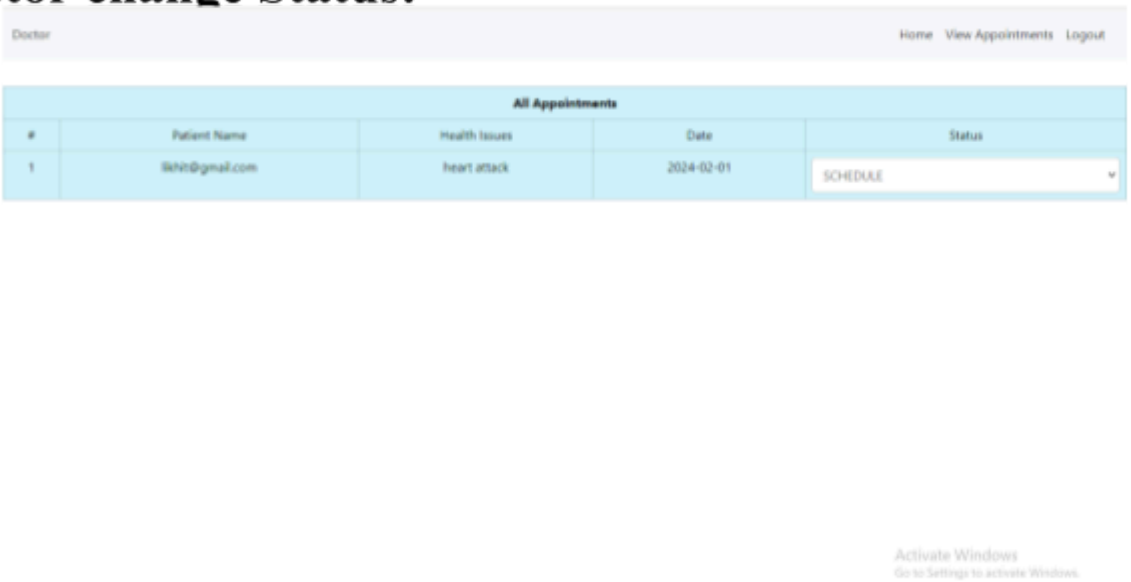
Doctor View Appointments:

All Appointments

#	Patient Name	Health Issues	Date	Status
1	lilchit@gmail.com	heart attack	2024-02-01	PENDING

Activate Windows
Go to Settings to activate Windows.

Doctor change Status:



Testing and Implementations

The purpose of testing is to discover errors. Testing is the process of trying to discover every conceivable fault or weakness in a work product. It provides a way to check the functionality of components, sub-assemblies, assemblies and/or a finished product It is the process of exercising software with the intent of ensuring that the
Software system meets its requirements and user expectations and does not fail in an unacceptable manner. There are various types of tests. Each test type addresses a specific testing requirement.

In testing process different levels of testing have been used as mentioned below.

1. Unit testing.
2. Integration question.

The overall testing process was accomplished through the following steps:

1. Unit testing

Unit testing involves the design of test cases that validate that the internal program logic is functioning properly, and that program inputs produce valid outputs.

All decision branches and internal code flow should be validated. It is the testing of individual software units of the application .it is done after the completion of an individual unit before integration. This is a structural testing, that relies on knowledge of its construction and is invasive.

Unit tests perform basic tests at component level and test a specific business process, application, and/or system configuration. Unit tests ensure that each unique path of a business process performs

accurately to the documented specifications and contains clearly defined inputs and expected results.

2. Integration testing

Integration tests are designed to test integrated software components to determine if they actually run as one program. Testing is event driven and is more concerned with the basic outcome of screens or fields. Integration tests demonstrate that although the components were individually satisfaction, as shown by successfully unit testing, the combination of components is correct and consistent. Integration testing is specifically aimed at exposing the problems that arise from the combination of components.

Software integration testing is the incremental integration testing of two or more integrated software components on a single platform to produce failures caused by interface defects.

The task of the integration test is to check that components or software applications, e.g. components in a software system or – one step up – software applications at the company level – interact without error.

Conclusion

In conclusion, this project presents a robust solution for optimizing doctor availability and appointment allocation in hospitals through a seamlessly integrated digital platform. By incorporating AI-powered features and fostering efficient patient-doctor communication, the system aims to revolutionize healthcare service delivery. With a focus on minimizing appointment numbers, reducing time slots, and providing real-time information, the project contributes to enhanced operational efficiency and improved patient experiences, marking a significant stride towards the advancement of digital healthcare.

Bibliography

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